# EduFin SQL Business Challenge

Phase 4 — Thursday Submission

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## STEP 4A — Open Metric Design: "Partner Health Score"

```
WITH inst_summary AS (
    SELECT
        i.institution_id,
        i.institution_name,
        COUNT(*) AS loan_count,
        ROUND(SUM(l.loan_amount) / 10000000.0, 2) AS loan_amount,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
        ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE
0 END) / 10000000.0, 2) AS default_amount
    FROM institutions i
    INNER JOIN loans l
        ON i.institution_id = l.institution_id
    GROUP BY i.institution_id, i.institution_name
)
SELECT
    institution_id AS `Institution ID`,
    institution_name AS `Institution Name`,
    loan_count AS `Loans`,
    CONCAT(loan_amount, ' Cr') AS `Loan Amount`,
    default_count AS `Default Count`,
    CONCAT(default_amount, ' Cr') AS `Default Amount`,
    ROUND(
        (default_count * 100 / loan_count) * 0.6
        + (default_amount * 100 / loan_amount) * 0.4,
        2
    ) AS health_score
FROM inst_summary
ORDER BY health_score DESC;
```

## STEP 4B — Default Rate Per Institution

```
WITH default_stats AS (
    SELECT
        i.institution_id,
        i.institution_name,
        COUNT(*) AS loan_count,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
        ROUND(default_count * 100.0 / loan_count, 2) AS default_rate
    FROM institutions i
    INNER JOIN loans l
        ON i.institution_id = l.institution_id
    GROUP BY i.institution_id, i.institution_name
)
SELECT
    institution_id AS `Institution ID`,
    institution_name AS `Institution Name`,
    loan_count AS `Loans`,
    default_count AS `Default Count`,
    CONCAT(default_rate, '%') AS `Default Rate (%)`
FROM default_stats
ORDER BY default_rate DESC, default_count DESC;
```

| Institution ID | Institution Name | Loans | Default Count | Default Rate (%) |
|---|---|---|---|---|
| 29 | State Commerce Institute | 10005 | 2088 | 20.87% |
| 33 | Central Engineering Institute | 9878 | 2050 | 20.75% |
| 34 | Modern Management Institute | 10165 | 2091 | 20.57% |
| 31 | Regional Commerce University | 9915 | 2039 | 20.56% |
| 30 | National Engineering Institute | 9889 | 2017 | 20.40% |
| 32 | National Commerce Institute | 9947 | 2028 | 20.39% |
| 50 | Government Medical College | 9780 | 1967 | 20.11% |
| 28 | Regional Medical University | 10134 | 1737 | 17.14% |
| 25 | Central Commerce College | 10009 | 1688 | 16.86% |
| 26 | State Arts & Science University | 10108 | 1687 | 16.69% |
| 24 | Modern Law College | 10151 | 1682 | 16.57% |
| 27 | Central Management College | 9984 | 1641 | 16.44% |
| 23 | Advanced Technology University | 10121 | 1652 | 16.32% |
| 36 | Indian Arts & Science University | 9897 | 1324 | 13.38% |
| 49 | Advanced Commerce University | 9926 | 1311 | 13.21% |
| 8 | State Engineering Institute | 9860 | 1300 | 13.18% |
| 13 | State Medical University | 10259 | 1330 | 12.96% |
| 38 | Central Management College | 10039 | 1293 | 12.88% |
| 12 | Central Commerce University | 10082 | 1291 | 12.80% |
| 21 | Central Law University | 9956 | 1273 | 12.79% |
| 46 | Regional Law College | 9886 | 1254 | 12.68% |
| 7 | Indian Law University | 10024 | 1270 | 12.67% |
| 48 | Central Commerce University | 9916 | 1254 | 12.65% |
| 4 | Advanced Arts & Science College | 10166 | 1282 | 12.61% |
| 2 | Central Law University | 10107 | 1274 | 12.61% |

## STEP 4C — Financial Loss Attribution

```
WITH financial_exposure AS (
    SELECT
        i.institution_id,
        i.institution_name,
        SUM(l.loan_amount) / 10000000.0 AS loan_amount,
        ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE
0 END) / 10000000.0, 2) AS default_amount
    FROM institutions i
    INNER JOIN loans l
        ON i.institution_id = l.institution_id
    GROUP BY i.institution_id, i.institution_name
)
SELECT
    institution_id AS `Institution ID`,
    institution_name AS `Institution Name`,
    CONCAT(ROUND(loan_amount, 2), ' Cr') AS `Loan Amount`,
    CONCAT(ROUND(default_amount, 2), ' Cr') AS `Default Amount`
FROM financial_exposure
ORDER BY default_amount DESC;
```

| Institution ID | Institution Name | Loan Amount | Default Amount |
|---|---|---|---|
| 34 | Modern Management Institute | 461.35 Cr | 88.54 Cr |
| 31 | Regional Commerce University | 448.56 Cr | 85.99 Cr |
| 30 | National Engineering Institute | 451.53 Cr | 85.15 Cr |
| 29 | State Commerce Institute | 446.11 Cr | 83.96 Cr |
| 33 | Central Engineering Institute | 447.13 Cr | 83.5 Cr |
| 32 | National Commerce Institute | 454.88 Cr | 82.81 Cr |
| 50 | Government Medical College | 437.3 Cr | 79.94 Cr |
| 28 | Regional Medical University | 454.79 Cr | 68.86 Cr |
| 26 | State Arts & Science University | 456.33 Cr | 68.38 Cr |
| 25 | Central Commerce College | 448.52 Cr | 66.67 Cr |
| 24 | Modern Law College | 450.19 Cr | 64.85 Cr |
| 27 | Central Management College | 443.59 Cr | 64.64 Cr |
| 23 | Advanced Technology University | 446.03 Cr | 63.76 Cr |
| 13 | State Medical University | 459.15 Cr | 51.87 Cr |
| 12 | Central Commerce University | 457.47 Cr | 51.42 Cr |
| 36 | Indian Arts & Science University | 448.59 Cr | 50.59 Cr |
| 38 | Central Management College | 457.73 Cr | 50.51 Cr |
| 7 | Indian Law University | 451.27 Cr | 49.63 Cr |
| 4 | Advanced Arts & Science College | 451.9 Cr | 49.19 Cr |
| 21 | Central Law University | 453.34 Cr | 49.09 Cr |
| 2 | Central Law University | 452.22 Cr | 48.92 Cr |
| 49 | Advanced Commerce University | 445.75 Cr | 48.64 Cr |
| 14 | Indian Technology University | 455.77 Cr | 48.56 Cr |
| 48 | Central Commerce University | 442.78 Cr | 48.51 Cr |
| 5 | Regional Law University | 440.85 Cr | 48.43 Cr |
| 41 | Indian Technology College | 445.59 Cr | 48.33 Cr |

## STEP 4D — Minimum Data Threshold Gate

```
WITH all_institutions AS (
    SELECT
        i.institution_id,
        i.institution_name,
        COUNT(l.loan_id) AS loan_count
    FROM institutions i
    LEFT JOIN loans l
        ON i.institution_id = l.institution_id
    GROUP BY i.institution_id, i.institution_name
)
SELECT
    COUNT(*) AS `Institution Count`,
    SUM(CASE WHEN loan_count < 50 THEN 1 ELSE 0 END) AS `Excluded (< 50 Loans)`,
    SUM(CASE WHEN loan_count >= 50 THEN 1 ELSE 0 END) AS `Included (>= 50 Loans)`
FROM all_institutions;
```

| Institution Count | Excluded (< 50 Loans) | Included (>= 50 Loans) |
|---|---|---|
| 5000 | 4950 | 50 |

Time: 0.528 seconds

## STEP 4E — Partnership Review Dashboard (Final Output)

```
WITH institution_summary AS (
    SELECT
        i.institution_id,
        i.institution_name,
        i.institution_type,
        COUNT(*) AS loan_count,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
        ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 /
NULLIF(COUNT(*), 0), 2) AS default_rate,
        ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE
0 END) / 10000000.0, 2) AS default_amount
    FROM institutions i
    LEFT JOIN loans l
        ON i.institution_id = l.institution_id
    GROUP BY i.institution_id, i.institution_name, i.institution_type
    HAVING COUNT(l.loan_id) > 50
),
flag_data AS (
    SELECT
        *,
        CASE
            WHEN default_rate >= 1.5 * (SELECT AVG(default_rate) FROM
institution_summary) THEN 'Suspend'
            WHEN default_rate >= (SELECT AVG(default_rate) FROM institution_summary)
THEN 'Review'
            WHEN default_rate >= 0.5 * (SELECT AVG(default_rate) FROM
institution_summary) THEN 'Retain'
            ELSE 'Reward'
        END AS partner_flag,
        CASE WHEN EXISTS (
            SELECT 1 FROM customers c
            JOIN loans l
                ON c.customer_id = l.customer_id
            WHERE l.institution_id = institution_summary.institution_id AND
c.cibil_score > 900
        )
        THEN 'Data Quality Review Required'
        ELSE 'Data Clean' END AS anomaly_flag
    FROM institution_summary
)
SELECT
    institution_id AS `Institution ID`,
    institution_name AS `Institution Name`,
    institution_type AS `Institution Type`,
    loan_count AS `Loans`,
    CONCAT(default_rate, '%') AS `Default Rate (%)`,
    CONCAT(default_amount, ' Cr') AS `Default Amount`,
    partner_flag AS `Partner Flag`,
    anomaly_flag AS `Anomaly Flag`
FROM flag_data
ORDER BY default_rate DESC;
```

| Institution ID | Institution Name | Institution Type | Loans | Default Rate (%) | Default Amount | Partner Flag | Anomaly Flag |
|---|---|---|---|---|---|---|---|
| 29 | State Commerce Institute | Institute | 10005 | 20.87% | 83.96 Cr | Suspend | Data Clean |
| 33 | Central Engineering Institute | Institute | 9878 | 20.75% | 83.5 Cr | Review | Data Quality Review Required |
| 34 | Modern Management Institute | Institute | 10165 | 20.57% | 88.54 Cr | Review | Data Clean |
| 31 | Regional Commerce University | University | 9915 | 20.56% | 85.99 Cr | Review | Data Clean |
| 30 | National Engineering Institute | Institute | 9889 | 20.40% | 85.15 Cr | Review | Data Quality Review Required |
| 32 | National Commerce Institute | Institute | 9947 | 20.39% | 82.81 Cr | Review | Data Quality Review Required |
| 50 | Government Medical College | College | 9780 | 20.11% | 79.94 Cr | Review | Data Clean |
| 28 | Regional Medical University | University | 10134 | 17.14% | 68.86 Cr | Review | Data Clean |
| 25 | Central Commerce College | College | 10009 | 16.86% | 66.67 Cr | Review | Data Clean |
| 26 | State Arts & Science University | University | 10108 | 16.69% | 68.38 Cr | Review | Data Quality Review Required |
| 24 | Modern Law College | College | 10151 | 16.57% | 64.85 Cr | Review | Data Clean |
| 27 | Central Management College | College | 9984 | 16.44% | 64.64 Cr | Review | Data Quality Review Required |
| 23 | Advanced Technology University | University | 10121 | 16.32% | 63.76 Cr | Review | Data Clean |
| 36 | Indian Arts & Science University | University | 9897 | 13.38% | 50.59 Cr | Retain | Data Clean |
| 49 | Advanced Commerce University | University | 9926 | 13.21% | 48.64 Cr | Retain | Data Clean |
| 8 | State Engineering Institute | Institute | 9860 | 13.18% | 47.59 Cr | Retain | Data Clean |
| 13 | State Medical University | University | 10259 | 12.96% | 51.87 Cr | Retain | Data Clean |
| 38 | Central Management College | College | 10039 | 12.88% | 50.51 Cr | Retain | Data Clean |
| 12 | Central Commerce University | University | 10082 | 12.80% | 51.42 Cr | Retain | Data Quality Review Required |
| 21 | Central Law University | University | 9956 | 12.79% | 49.09 Cr | Retain | Data Clean |
| 46 | Regional Law College | College | 9886 | 12.68% | 47.21 Cr | Retain | Data Clean |
| 7 | Indian Law University | University | 10024 | 12.67% | 49.63 Cr | Retain | Data Clean |
| 48 | Central Commerce University | University | 9916 | 12.65% | 48.51 Cr | Retain | Data Clean |
| 2 | Central Law University | University | 10107 | 12.61% | 48.92 Cr | Retain | Data Clean |
| 4 | Advanced Arts & Science College | College | 10166 | 12.61% | 49.19 Cr | Retain | Data Quality Review Required |
| 40 | Central Medical University | University | 10043 | 12.60% | 47.03 Cr | Retain | Data Quality Review Required |
| 15 | Premier Management College | College | 9793 | 12.55% | 46.9 Cr | Retain | Data Clean |

## Custom Column

```
CASE
    WHEN default_rate >= 1.5 * (SELECT AVG(default_rate) FROM institution_summary)
THEN 'Suspend'
    WHEN default_rate >= (SELECT AVG(default_rate) FROM institution_summary) THEN
'Review'
    WHEN default_rate >= 0.5 * (SELECT AVG(default_rate) FROM institution_summary)
THEN 'Reward'
    ELSE 'Retain'
END AS partner_flag

CASE WHEN EXISTS (
    SELECT 1 FROM customers c
    JOIN loans l
        ON c.customer_id = l.customer_id
    WHERE l.institution_id = institution_summary.institution_id AND c.cibil_score >
900
)
THEN 'Data Quality Review Required'
ELSE 'Data Clean' END AS anomaly_flag
```

> A fixed threshold like default_rate > 15% assumes you already know what "bad" looks like before seeing the data, but that number is arbitrary and breaks down across different economic conditions, loan types, or time periods. Comparing against the portfolio average is self-calibrating: if the whole portfolio is performing poorly (say average is 18%), a 15% institution is actually your best partner, not a suspension candidate. It also means your flags automatically adapt as the portfolio changes — no one needs to manually update hardcoded thresholds every quarter.

## WHY Gate

**1. The institution with 0 defaults — what action did you assign it, and why? Write your classification AND your reasoning. ("I assigned it X because Y, not Z because A.")**

'National Law Institute' (Institution ID 17) is the only institution with 0 defaults. I classified the institution as 'Reward' and not 'Retain', 'Review' OR 'Suspend', because its default rate of 0.00% falls below 0.5x the portfolio average, making it the strongest performer/institution in the entire dataset. I did not classify it as 'Retain' because Retain is reserved for institutions between 0.5x and 1x the average, which still carry moderate risk. 'Review' and 'Suspend' are ruled out entirely since those require a default rate at or above the portfolio average.

**2. Write your Suspension Justification for the top-ranked institution to be suspended. Format it as if going to a legal review: Institution name, default rate, loan count, total defaulted exposure, which threshold was exceeded, recommended action, and what monitoring must happen post-suspension.**

- Institution: State Commerce Institute (Institution ID: 29) - Default Rate: 20.87% - Total Loans Disbursed: 10005 - Total Defaulted Exposure: 83.96 Crores - Threshold Exceeded: 20.87% exceeds the 1.5x portfolio average threshold - Recommended Action: Immediate suspension of new loan disbursements pending full portfolio audit. No new student loan applications to be processed through this institution until review is complete. - Post-Suspension Monitoring Requirements: - 83.96 Cr in defaulted exposure must be assigned to collections with a 90-day recovery report due to the Partnership Director - Institution returned Data Clean on the anomaly check, meaning no CIBIL >900 irregularities were detected; however a manual data audit is still recommended given the volume - Partnership agreement to be placed under legal hold; no contract renewals or amendments permitted during suspension period.

**3. If you were the Partnership Director and received your final output — what is the FIRST call you make? Who do you call, about which institution, and what specifically do you say in that call?**

If i were the partnership director that receive my own final output, then i would first try to call the institution who had 0% default rate despite having 10026 loans. This result did not came from an rouding error, but due to the structural difference of how you guys are operating compared to the other institutions. Before I spend the next few weeks on suspension hearings for the bottom of this portfolio, I want to understand what you're doing differently like, what's your screening process, your student selection criteria, your collections touchpoints, whatever it is. Because if we can document it and replicate it, that's worth more to this organization than suspending 10 bad partners.